



Karachi AQI Hub

AI-Powered Air Quality Forecasting & Health Intelligence

System Status

Champion Model: Gradient Boosting

Test R² Score: 0.9113

Test RMSE: 2.39 µg/m³

Hopsworks Feature Store

Feature Group:

karachi_aqi_features (v1)

Selected Features: 18 Predictors

Sync Mode: Hourly GitHub Actions Cron

Refresh Live Data

Built for 10 Pearls Data Science Internship • Karachi, Pakistan

Karachi Air Quality Index & 72-Hour Forecaster

Karachi, Pakistan • Real-Time Satellite & Atmospheric Telemetry • Updated: Thursday, August 27, 2026 • 09:00 AM UTC

LIVE AQI STATUS: MODERATE



PM10 Coarse Dust

62.4 µg/m³

Temperature

29.1 °C

Relative Humidity

70.0 %

Wind Speed

19.4 km/h

Carbon Monoxide (CO)

193 µg/m³

Surface Pressure

1002 hPa

Health Advisory & Action Plan: Air quality is acceptable; moderate health concern for sensitive individuals.

72-Hour (3-Day) AQI Forecast | Exploratory Data Analysis & Trends | SHAP Explainability & Factor Insights | Scenario Simulator Sandbox | Model Leaderboard & MLOps Registry

Model Interpretability & SHAP Feature Attribution

Understand precisely why the machine learning model made its current PM2.5 prediction.

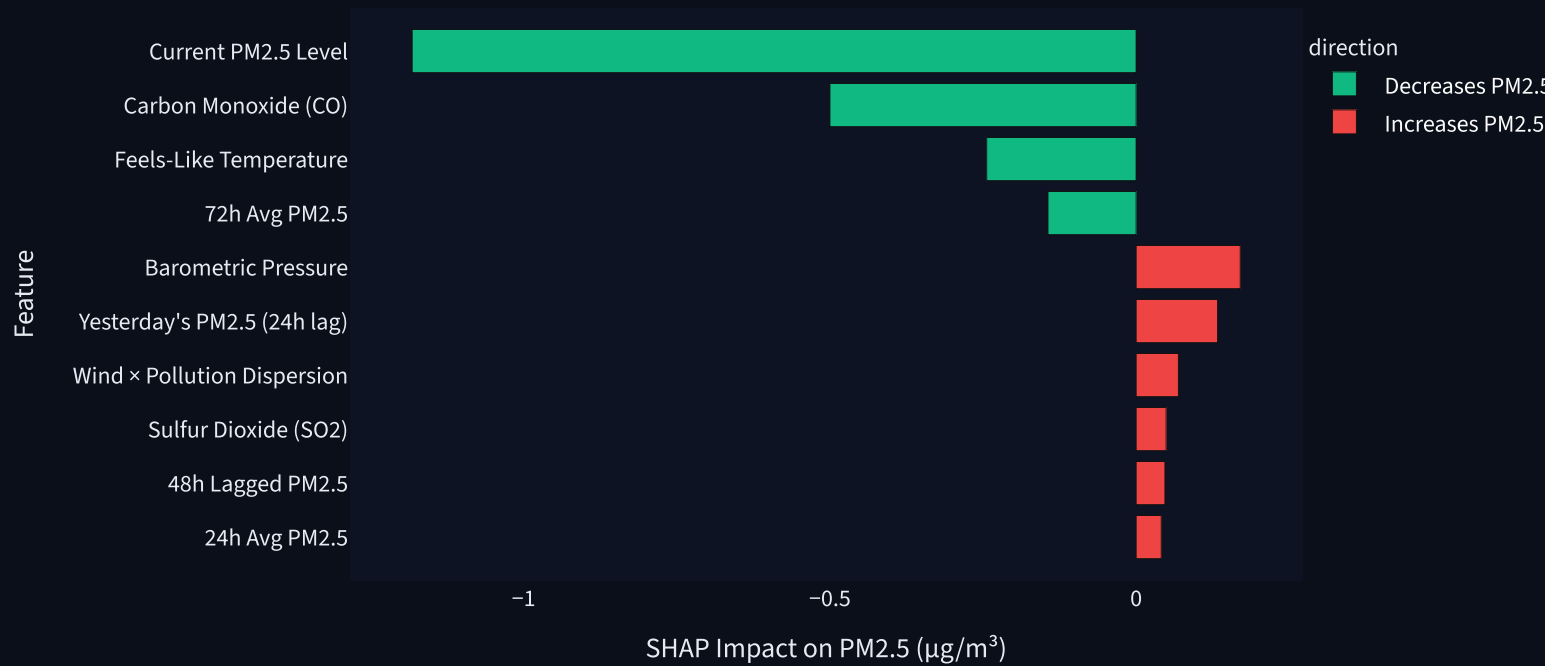
The AI model predicts a PM2.5 concentration of 27.9 µg/m³ (baseline expectation is 29.5 µg/m³).

Key factors elevating pollution: Barometric Pressure (+0.17), Yesterday's PM2.5 (24h lag) (+0.13).

Key factors mitigating pollution: Current PM2.5 Level (-1.18), Carbon Monoxide (CO) (-0.50).

Local Feature Impact for Latest Prediction

Top 10 Feature Contributions to Current AQI



Global Feature Importance Ranking

Global Model Feature Importance (%)

